

URPro-B Series (Outdoor) User Manual

Preface

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Guarantee and Compensation

We provide a guarantee relating to perfect manufacturing as part of the legally stipulated terms of guarantee. On receipt, the purchaser must immediately inspect all delivered goods for damage incurred during transport, as well as for material and manufacturing faults UNIVIEW must be informed immediately in writing of any complaints.

The environmental conditions as well as the servicing and maintenance regulations specified in the this manual must be complied with by the customer.

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I .Safety information



Warning:
Make sure to understand and follow all of the manual's safety information, important warnings and notes.

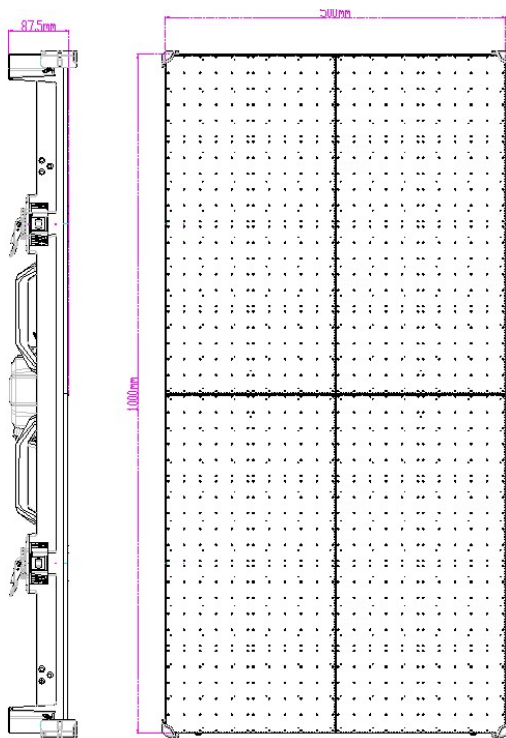
- ▶ Don't install the product close to heat sources like radiators, heaters, stoves or other equipment that can produce heat (including amplifiers).
 - ▶ Don't tread power and data cables; especially avoid squeezing plugs, outlets and power/data cables led from equipment. Replace damaged power and data cables immediately.
 - ▶ Only the company designated accessories and fittings can be used.
 - ▶ Please cut off the power or provide other applicable lightning protection during a lightning storm. Please cut off the power if the product is not on use for a long time.
 - ▶ The mains plug of appliance coupling is used as the disconnected device and the socket-outlet shall be installed near the equipment and shall be easily accessible.
 - ▶ The product shall be installed and maintained by qualified tech service staff. Please contact professional maintenance personnel to repair the product damaged due to various reasons (such as power or plug broken down, equipment failure or falling down).
 - ▶ Be careful when uplifting, moving or transporting the product, avoiding personal injury and product damage due to falling down.
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- ▶ Improper operations may result in electric shock, fire or falling down, causing serious damage or death.
 - ▶ The input voltage of the product is divided into: AC 100-240V 50/60Hz in accordance with different areas.
 - ▶ When performing installation and wiring, make sure to turn off the master switch of input power of the whole display, and the product must be properly grounded.
 - ▶ Do not provide the product output power terminals for other equipment. Carefully check if there is trouble in the power cable and its joint and terminal blocks before powering on.
 - ▶ Don't use any damaged or bare power cables or power plugs.
-
- ▶ Please provide at least 30cm space for heat dissipation of the rear radiator.
 - ▶ Don't directly place or paste any articles on the lamps.
 - ▶ As the product generates quantities of heat, please put the product far away from inflammable and explosive objects. Ensure adequate ventilation during installation for safe running of the product.
 - ▶ To avoid causing fire due to the overloaded power cables, please properly allocate the load of power cables according to this user manual or use cables configured by UNIVIEW LED.
 - ▶ Pay attention to the hanging load; perform hanging installation in strict accordance with the load limit.
 - ▶ Please wear helmets during construction to avoid causing personal injury.
 - ▶ Please fasten the seat belt when working high above the ground.

II .Introduction URPro-B(Outdoor) LED Display

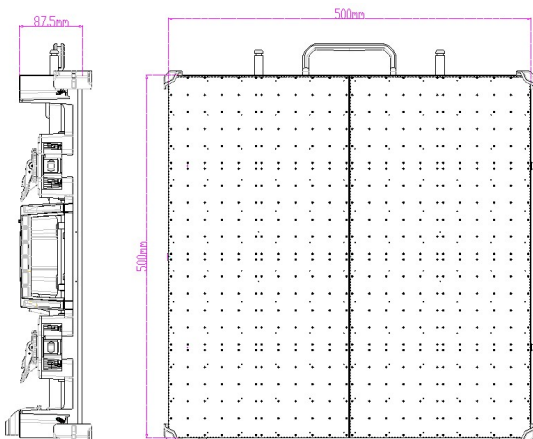
Description:

- ▶ Easy Set-up and installation.
- ▶ High assembling precision, fast lock design, quick and easy disassemble.
- ▶ Universal led modules can be used in different size cabinets.
- ▶ Low gray and high refresh rate:1 level(0-255 levels) gray scale, reachable refresh 15360Hz.
- ▶ High contrast ratio:16Bit high precision gray scale delivers delicate image quality.

Cabinet Size:500*1000*87.5mm



Cabinet Size:500*500*87.5mm

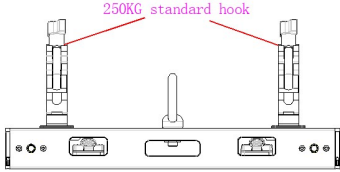
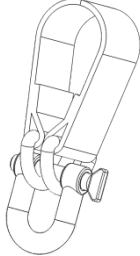
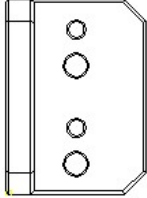


Product Technical parameters

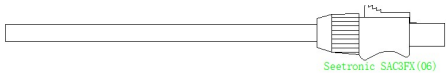
URPro-B(Outdoor) Technical parameters	
Pitch Pixel	3.9
Pixel composition	SMD1921
Applicable environment	outdoor
Module dimensions (mm)	250*500
Module resolution	64*128
Luminance (cd/m ²)	3000-3500/4000-4500/5000
Gray scale (bit)	≥16bits
Refresh rate (Hz)	≥15360Hz
Power Consumption (Max./Avg.)(W/m ²)	750/248
Cabinet Size(mm)	500×500x87.5mm / 500×1000x87.5mm
Input Voltage (V)	100-240V
Viewing Angle	H : 140° V : 140°
Temperature Range (°C)	Operation:-20°C ~ +50°C
Service Life	100,000H
IP Rating (Front/Rear)	IP65/IP55
Maintenance	Front and rear
Installation	Stacking / Hanging

III. LED installation and Maintenance

1, Installation Accessories and Tools

		
Hanging beam (Type A)- Use for hanging or stacking	Hanging beam (Type B) - Use for hanging installation only.	Hoisting suite — With hanging bar for hanging cabinet.
		
Hexagonal wrench – Use to fix or loosen screws	Connecting piece- Use for corner screen installation.	Front maintenance tool -Use for module maintenance

2, Cable

	Name: Main power cable Specifications: 3*2.5mm² multiple copper cores Length: 10/15m Color: black
	Name: Main signal cable Specifications: Length: 15m Color: Black/Yellow
	Name: Power cable between cabinets Specifications: The power connector can be cascaded with other cabinets. Length: 500mm Color: Black
	Name: Signal cable between cabinets Specifications: The signal connector can be cascaded with other cabinets. Length: 650/1300mm Color: Black/Yellow

3, Installation method

Note:

1. Please operate carefully to avoid the danger of falling of modules and other accessories.
2. Do not perform the following operations without disconnecting the power supply to the display

3.1 Hanging installation

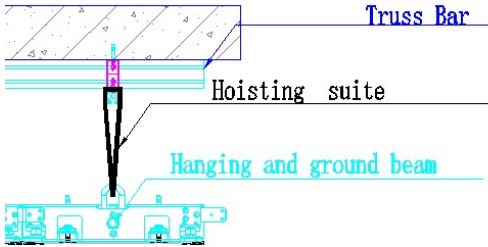
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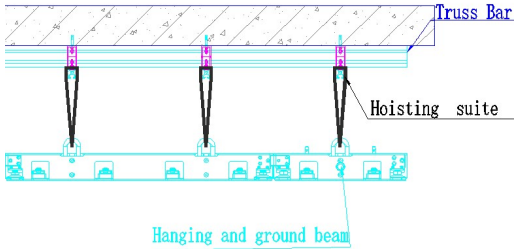
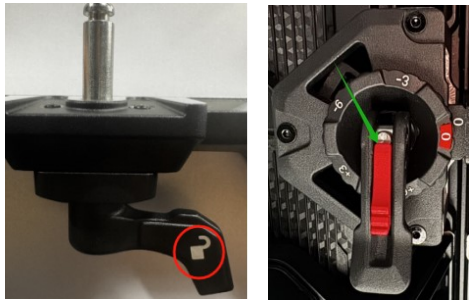
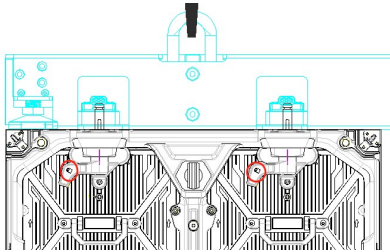
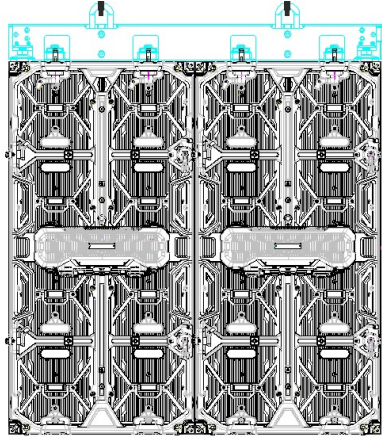
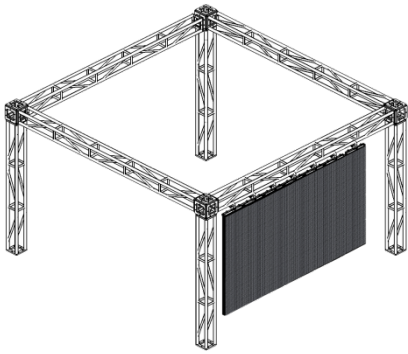
1. When this series is hanging, there are two types of hanging beams to choose from.
2. When the tensile force is 8 times the safety factor, the height of the LED display lifted by a single hanging beam shall not exceed 10m.
3. When the tensile force is 10 times the safety factor, the height of the LED display lifted by a single hanging beam shall not exceed 8m.
4. For curve hanging installation, only 500mm trimmed hanging beams can be used.

- ① Fix the required structure according to design requirements.
- ② Select the appropriate hanging beam according to the actual needs of the project.

	Name: Simple hanging beam Specifications: Only suitable for hanging installation. Length: 500/1000mm Color: Black
	Name: Hanging and stacking beam Specifications: It can be used as a hanging beam or with an installation frame as an installation base. Length: 500/1000mm Color: Black

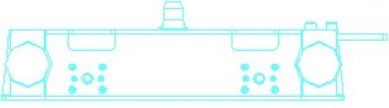
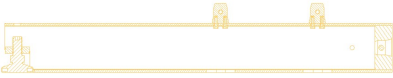
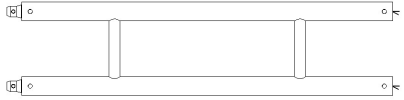
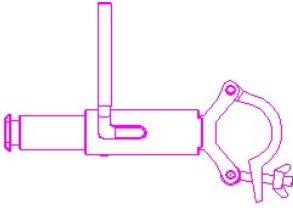
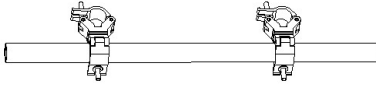
- ③ Here is an example of a single hanging beam.

	► Fix the hanging beam with truss bar according to design requirements. And adjust the flatness of surface of the hanging beams.
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 Truss Bar Hoisting suite Hanging and ground beam	► Adjust the distance and flatness between hanging beams. Then use an Allen wrench to fix the hanging beams through the built-in connecting pieces.
	► Adjust the handle of the top lock to be perpendicular to the horizontal plane of the cabinet. ► The handle of the lock is rotated 90 degree to the right, the lock lever can be fixed first. ► Press the switch on the arc-shaped lock on the right side of the frame, open the side lock and adjust it to the corresponding angle, and then lock the switch again.
	► Pick up the cabinet and insert the lock into the connector with the hanging beam. After the flatness of the hanging beam and lower cabinet is adjusted, continue to rotate the lock handle to the left, and then fix the connection between them.
	► Fix another cabinet to the hanging beam follow above steps. ► After the flatness between the cabinets is adjusted, use the left and right side locks of the frame to lock them tightly.
	► According to the above steps, complete the installation of the display screen.

3.2 Stacking installation

3.2.1 Installation Accessories

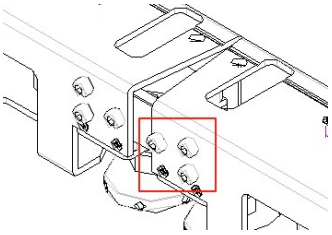
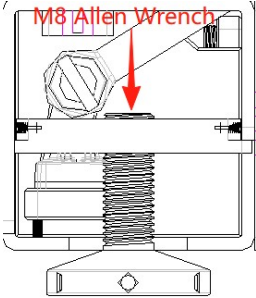
		
Hanging and stacking beam	Base beam	Ladder Truss
		
Back frame connector	Ladder connecting rod	

3.2.2 Installation Method

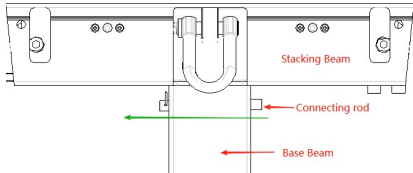
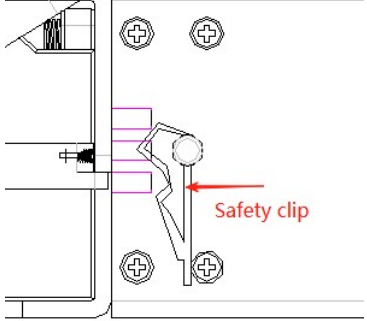
Note:

1. When the display screens are stacked, additional reinforcement methods must be added to the display screens to prevent the display screens from falling over.
2. Our company will not bear any responsibility for all losses caused by failure to operate as required.
3. For curve stacking installation, only 500mm trimmed hanging beams can be used.

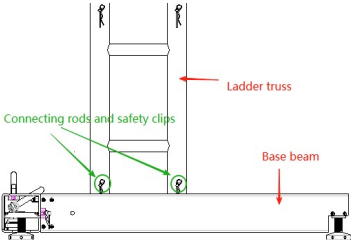
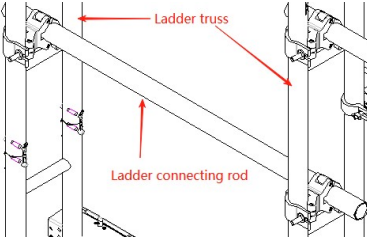
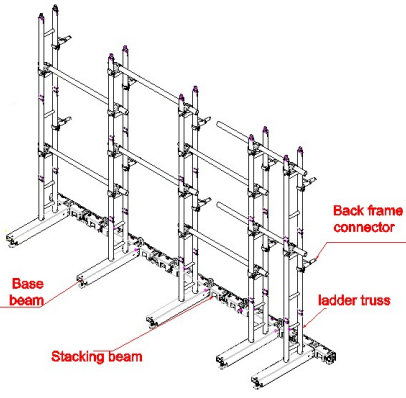
- ① Prepare all the required accessories according to design requirements.
- ② Connect the hanging and stacking beam together by tightening the screws.

	► Tighten the screws using a M6 Allen wrench.
	► If the beams are not level, rotate the feet to adjust the height using a M8 Allen wrench.

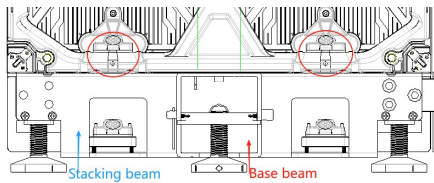
- 3 Connect the base beam with the hanging and stacking beam with a connecting rod and a safety clip.

	► Connect them together using connecting rods on the base beam. ► The connecting rod is inserted from right to left
	► Insert a safety clip to prevent connecting rod from falling off.

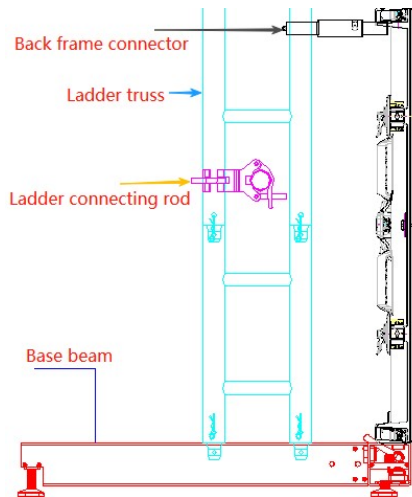
- 4 Mount the ladder truss on the base beam and connect them with connecting rods and safety clips

	► Put the ladder truss on the base, Connect them together using connecting rods on the base beam. ► The connecting rod is inserted from right to left.
	► Use ladder connecting rod to connect every two ladder trusses.
	► Strengthen the entire structure. For example: fixing the Base beam to the ground. ► According to the above steps, complete the installation of the stacking structure.

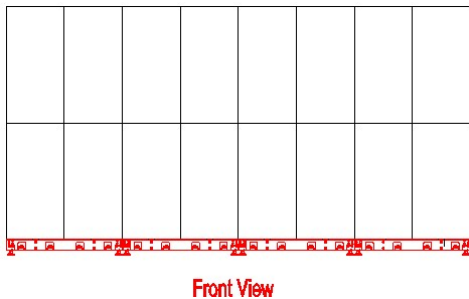
5 Install the cabinets on the stacking beams.



- Pick a cabinet and insert the bottom lock socket into the connector with the hanging beam.



- Adjust the flatness between the cabinet and the stacking beam.
 - Use the 'Back frame connector' to secure the cabinet to the "Ladder Truss" behind it.
- Note: The connection location of this series is at the top of the frame.

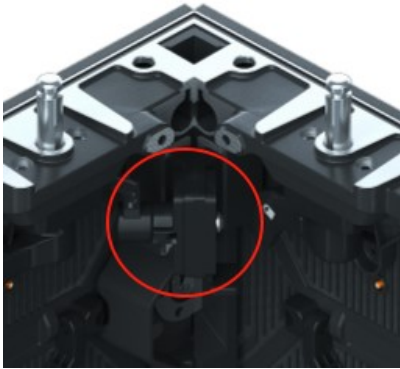
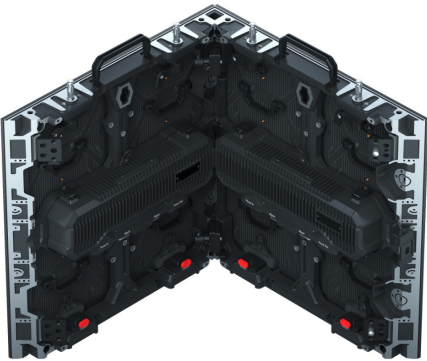


- According to the above steps, complete the installation of the display screen.

3.3 Corner installation

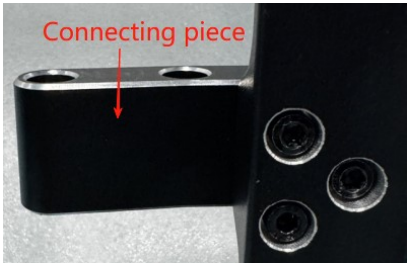
Note: Only the 45 degree trimmed cabinets of 500*500mm can be assembled into a corner screen.

3.3.1 Installation Method

	▶Align the trimmed frame. ▶Use the installed left and right 45° quick locks on the frame and pass through the corresponding lock holes from left to right. ▶Turn the handle clockwise to lock the left and right cabinets tightly.
	▶Lock the quick lock on the bottom of the frame according to the above method. ▶Complete the corner installation between cabinets.

3.4 How to use windproof frame with cabinet.

Note: The windproof frame needs to be used with 500*1000 cabinets.

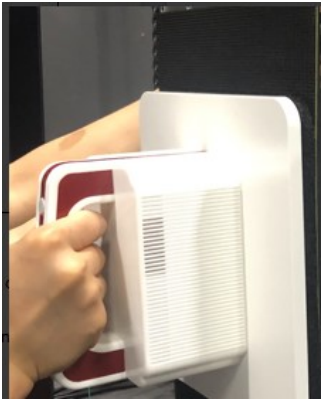
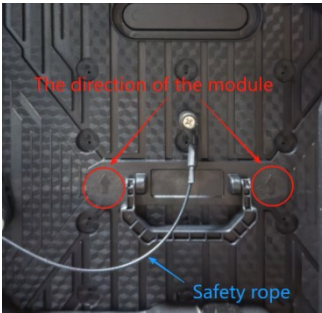
	► Use M5*30 hexagon screws to fix the connecting piece connecting the windproof frame at the corresponding position on the cabinet frame. ► Each cabinet frame has a total of 4 connecting pieces that need to be installed.
	► Prepare the windproof frame that need to be installed. ► Remove the latch on the windproof frame from the lock hole. Note: We need to press and hold the button on the top of the latch to remove the latch from the lock hole.
	► Align the latch holes of the windproof frame with the outer mounting holes on the connection pieces. ► Press and hold the button on the top of the latch and pass through the installation hole to complete the fixation between the windproof frame and the cabinet frame.
	► The effect after installation is completed is as shown in the figure.

4, Maintenance method

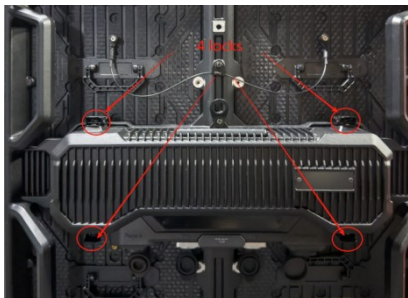
Note:

1. Please operate carefully to avoid the danger of falling of modules and other accessories.
2. Do not perform the following operations without disconnecting the power supply to the display

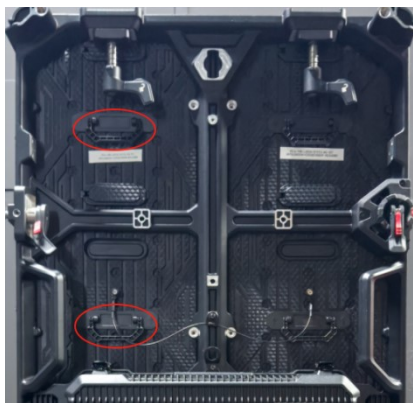
4.1 How to maintain the module from front.

	▶ Prepare the maintenance tool in advance.
	▶ Turn on the power of maintenance tool. ▶ Place the maintenance tool carefully on the surface of the module and attach it completely. ▶ Press the white button on the handle of the maintenance tool to make it work properly.
	▶ After holding the module stabilizer by hand, press the white button on the maintenance tool again to close it. Then loosen the safety rope connecting one end of the module. ▶ Carefully remove the module parallel to the ground to avoid secondary physical damage.
	Install modules back. ▶ Before installing the module, keep the direction of the module consistent with the direction of the cabinet frame. ▶ Use safety ropes to reconnect the cabinet frame and the corresponding positions of the module. Make sure the module does not fall. ▶ When installing the module, the connection socket between the module and the control box must be inserted horizontally to prevent damage to the module's related accessories.

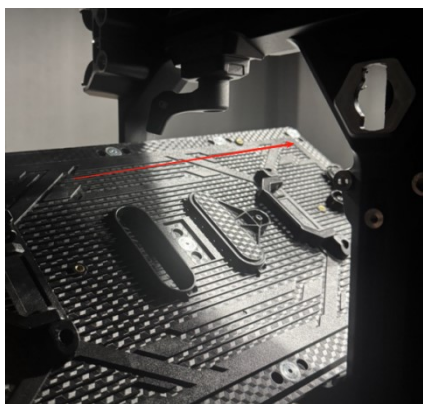
4.2 How to maintain the module or power box from rear.



- ▶ The power box can be opened directly from rear maintenance.
- ▶ We only need to open the 4 locks as shown in the figure to replace a complete power box..



- ▶ Hold the upper and lower handles on the cover of the defective module .
- ▶ Push the module out forcibly in the direction perpendicular to the box frame.
- ▶ Loosen the safety rope connecting one end of the module.



Install modules back.

- ▶ Before installing the module, keep the direction of the module consistent with the direction of the cabinet frame.
- ▶ Pass the spare modules through the frame as shown in the figure, and adjust the position and direction of the modules.
- ▶ When installing the module, the connection socket between the module and the control box must be inserted horizontally to prevent damage to the module's related accessories.
- ▶ Use safety ropes to reconnect the cabinet frame and the corresponding positions of the module. Make sure the module does not fall.

IV. Common Faults of LED Display and Solutions

Fault	Causes	Solutions
Black screen	1. No power supply to the display. 2. Signal transmission cable is not well connected. 3. The graphics card of computer is not well placed. 4. The screen is turned off by the control software. 5. There's trouble with the control system. 6. The receiving card is damaged. 7. The control system has no display program parameters.	1. Make sure the screen is properly powered. 2. Check whether the DVI cable and other transmission cables are well connected. 3. Set the graphics card of computer as copy or expansion mode. 4. Set the control system software as the state of display open. 5. Check whether the control system is well installed. The green light blinking and red light on means normal operating; the green light on or off means receiving card broken or having no signal, and replace if it is damaged in fact. 6. Check whether the receiving card works normally (red light on) and transmits signals (green light blinking) normally. 7. Open the control software to load the program parameters of display and send them to the receiving card of display for saving.
Dislocation or repetition of display picture	1. Incorrect configuration files of LED display. 2. Connection error of the signal cable between cabinets.	1. Reset the parameters of LED display. Reset the connection files of display according to the direction of signal cable and send them for saving. 2. Check whether the signal cable between cabinets is well connected and if there is wrong connection.
No lighting of single cabinet	1. Damaged power 2. Broken receiving card	1. Check whether cabinet is properly powered. 2. Check whether the receiving card is working (red light on) and transmits signals (green light blinking) normally. Please replace the receiving card if the green light is on or off.

V. Maintenance

Note: Please pay attention to the use methods and maintenance of LED display, so as to ensure a longer service life of it.

[I] Routine Maintenance of Main Control Computer

The computer is usually placed in the control room, thus to keep the stability and good displaying effect, we must maintain the main control computer regularly as follows:

1. Clean

Keep the control room clean and regularly dust the computer and surrounding equipment, as dust may contaminate the keyboard, disc, display, host circuit board and other important parts, causing material faults.

2. Ventilation

The computer will produce a lot of heat during operation. Poor heat dissipation and excessively high room temperature may lead to computer crash and other faults, and the service life of the main control computer may be reduced in a long term. Therefore, ventilation must draw enough attention.

3. Temperature and relative humidity

If the environmental temperature is relatively high (e.g. in hot summer), the control room must be equipped with fans or air condition equipment if applicable, so as to maintain the temperature as 10°C ~ 35°C and the relative humidity as 30% - 80%, and try to avoid using computer under high temperature and high humidity.

4. Hardware examination

Regularly examine the hardware and surrounding accessory equipment of the computer, e.g. whether the graphics card and sending card is loosened, whether DVI cable, serial port cable or USB cable is loosened or damaged, and eliminate discovered problems in time.

5. Software upgrading

Kill virus regularly, especially that the computers connecting internet shall be installed with anti-virus software. And the player software shall be upgraded regularly to assure the latest version is used.

6. Points for attention:

The more stable the computer power is, the better. Usually it is required to use AC 220 V $\pm$ 10% with frequency of 50HZ $\pm$ 5%. A fluctuate power can affect the normal running of computer and directly affect the regular displaying of LED display, thus it is suggested to add one UPS, if possible, which will protect the computer data from missing due to sudden blackout of main supply and also has voltage stabilization function generally.

If there are high-power refrigerators or air conditioners near the computer, the power socket of the computer shall be separated with theirs and shall be used independently.

As safety assurance, the computer power shall have good grounding system with three-phase power socket of neutral wire on the left, live wire on the right and ground wire on the top. The ground wire here is the safety wire of power, which should not be connected with the logical ground wire of the computer.

[II] Maintenance of LED Display

(I) Cabinet part

1.The examination of cabinet part mainly includes the following procedures:

Check whether the power supply works normally and whether AC wire is loosened.

Check whether the fan works normally, and dust the fan blade.

Check the receiving card: check whether the supplying DCSV wire is loosened and whether the receiving card and hub card are well contacted as well as whether the winding displacement of hub card is loosened, Check whether the DCSV wires and winding displacements between the internal modules of the cabinet are loosened.

Check whether there are sundries jamming the air-vent, and dismantle the anti-dust net to clear off if so.

(II) Display part

Regularly dust the surface of LED display especially for outdoor LED display which is long-time exposed to the work environment of wind, sun and dust and can be easily contaminated. Thus, regular dusting of LED display can not only elevate displaying effect but also avoid the acid particles and other chemical substances in the outdoor environment to corrode the surface of LED display under humid condition, so as to improve the service life of LED display.

For outdoor LED display, install efficient lightning-protection equipment if necessary. The following issues shall be paid attention to for application and maintenance of LED display:

1. It is required to have stable power supply and good grounding protection.
2. Don't use the LED display under bad weather, e.g. during lightning storm.
3. Water and iron powders and other conductive metals are forbidden to enter into the display. LED display shall be placed in low dust environment, as large dusts may affect the displaying effects and too much dusts will damage the circuits.
4. In case of water entering due to any reason, please switch off and contact the maintenance staff. The display shall not be used until the internal display board is dried. Maintain the humidity of the use environment of full-color LED display and do not let any item of humid nature to enter into the display. Powering on the full-color LED display with humidity can result in corrosion of parts, thus causing permanent damages.
5. The start and close sequence of LED display: A. First start the control computer to make it work normally, then start the LED display. B. First close the LED display, then computer.
6. LED display surface can be wiped by alcohol or be dusted by brush and dust collector instead of being wiped by wet cloth. In order to avoid possible problems, we can choose passive or positive protection by trying to keep the harmful items far away from the display and slightly wipe the display while cleaning so as to minimize the possibility of damage.
7. It is suggested that LED display shall rest for more than 2 hours each day and be used for at least once a week in rainy seasons. The display shall be started up for at least once a month and be on for more than 2 hours generally.
8. LED display shall be regularly examined to ensure the normal running and repair in time if there's fault; and examine if the circuit is damaged and repair or replace it if so.
9. Don't make LED display in long-time displaying of full white.
10. The relevant equipment such as main control computer of LED display shall be placed in a room with air conditioner and slight dusts, to ensure well ventilation and radiation and stable working of computer.
11. Non-professionals are not allowed to touch the internal circuits of LED display in case of electric shock or damage of circuit. Ask the professionals to examine and repair in case of problems.

[III] Storage of LED Display

(I) Preservation of the non-installed display, standby modules and other accessories

1. Perfect moisture-proof work at the storage places: appropriate environmental temperature of $-40^{\circ}\text{C} \sim +85^{\circ}\text{C}$ and relative humidity lower than 90%.
2. Prevent insects and mice from biting the display and cables during storage.
3. Prevent rainwater and other corrosive liquids from entering into or contacting the display and modules during storage.
4. Perfect the cleaning work during storage to avoid sundries or dusts piling up in the display.

(II) Usage and storage of LED display under special environment.

1. The display shall be avoided to be installed at waterlogged places. The power of display and computer shall be cut off in time in case of extreme severe convection weathers such as typhoon, storm, hail and thunder.
2. The cabinet and external steel frame shall be well grounded in case of display burnout, even human injury or death caused by lightning or display leakage.
3. For the displays which are installed in coastal areas, please pay attention to the corrosion of metal parts of display by air salts, as well as the anti-typhoon ability of the installed steel structure.

4. The displays in some countries and areas may not be able to work normally due to very low temperature in winter, thus it is suggested to use low-temperature or super low-temperature power and low temperature resistant cabinet, and warm air curtain can be added if necessary.

VI. After-sales Service

1. Provide free support of service line and email address

In case your maintenance personnel come across any maintaining or operating problems, you can call our after-sales service office to consult our technical experts for the relevant problems. Our service line operates 8 hours/day from 9:00 a.m. to 18:00 p.m. No matter what condition our service line works in, you can contact us by the 24-hour working email address for technical support by our Engineering and Service Department. Once we receive your email, we will try our best to answer your questions and solve relevant problems as fast as we can.

2. Provide free remote assistance service

In case the relevant technical problems feedback by you may not be solved by phone calls or emails in real-time, as the case may be, we will provide remote assistance or dispatch our engineers to the engineering site to help you.

3. Provide lifelong maintenance service of the product

Within the maintenance period, we will take charge of maintenance of all the equipment under the contract and replace or repair all the damaged equipment or parts due to product design, installation process, materials, product quality and accessories.

After the warranty period, In case of faults of module, power supply or control system, you may send it back to us directly for examination and repair. Our technical staff will return it to you in time once it is surely fixed. For the cases that couldn't be fixed, we will provide you with the same quality accessories, charging costs only.

4. Free upgrading

We will offer lifelong software upgrading of your equipment to provide the latest version of control software in time.

5. Other terms and conditions

The maintenance service is valid only for the originally provided equipment and original installation places and users. If the relevant equipment is re-installed in other places or the user transferred the ownership of the display equipment under the contract, the user shall inform us in written for re-assessment of relevant services. Otherwise, the relevant maintenance service will be invalid.

6. Long-time provision of spare products, parts and special tools

Within the service terms and conditions, we will store sufficient quantity of the same pipe core and components of the same production batch to ensure the quality of after-sales service so that the purchasers will not delay system operating due to lacking parts; we can provide equipment of similar types for temporary and alternate usage under special situations, so as to assure the maintenance service of the project is smoothly carried out.

7. Points for attention

Our service exclude the faults or damages that are caused by negligence, accidents, natural disasters, operating not complying to the user manual, misuse and handling products not in accordance with instructions, as well as self-modified products or missing parts.

Contact Us

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